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Lab Assignment 3 :

Objective : To write a lex script for a subset of C, generate a scanner and use it to find all tokens present in various input C source programs. The tokens are incrementally enhanced in order to help you write more interesting lex scripts and process C programs that are closer to the real programs that you would have written. *In case you find that the files given have errors, call the instructor(s) and inform, but in any case fix the errors and proceed with the experimentation.*

Find answer to the following question : Does the ordering of the tokens in the “rules section” make any difference to the performance of the generated scanner ?

P1. You are given a lex script named as “numregex1.l”. Read the 5 regex definitions given in this file, which include, regexes for base 8 and base 10 numbers, among others.

1.1 Generate a scanner for the given script. Report whether lex or gcc reported any errors or not with reasons. Run the generated scanner over the input file, “numinp1.txt” and report your observations with reasons.

1.2. Make changes to “numregex1.l” and name it as “numregex2.l”. When a scanner is generated from “numregex2.l”, and run on “numinp1.txt”, the output generated should be same as given in the file, “numout2.txt”.

1.3 Create a new lex script, say “numregex3.l”, by interchanging the order of the two tokens, {number} and {octalnum} given in the “rules section”. Repeat the steps described in 1.2 above and report your findings. Compare the outputs of scanners generated in 1.2 and 1.3 and explain the difference in behaviour, if any.

Resources given : Three files : “numregex1.l”, “numinp1.txt” and “numout2.txt”

P2. Extend the lex specifications to recognize the following additional tokens : hexadecimal numbers, real numbers (with and without the exponent form), character literals and a bitstring. A bitstring is a new numeric value introduced by us which is not part of C. A bitstring is a base 2 binary string that starts with “0b” or “0B”, followed by one of symbols from {0,1}. You are given an input file, “numinp3.txt” and the desired output of the recognizer is provided in “numout3.txt”. Read the desired output of each token type carefully and make sure that the regexes with the actions written by you produce the same display.

Resources given : Two files : “numinp3.txt” and “numout3.txt”

P3. You are required to add more features, such as recognize single line comment, recognize string literals (enclosed in double quotes), operators and more keywords to your lex specification. The resulting lex script should be able to recognize a reasonably good subset of C.

3.1 Read the program, “sampleprog.c” for the lexical features of C it uses. Compile and run the executable and examine the output and check its correctness.

3.2 You have to write a lex script that can recognize all the lexical elements of “sampleprog.c” and the scanner generated by you has to produce output as given in “sampleout.txt”.

Resources given : Two files, “sampleprog.c” and “sampleout.txt”

End of Lab Assignment 3